Homework 7

ORF 245 - Fundamentals of Statistics

Due: Apr. 11th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
- Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).

Exercise: 1. (Simulation: CI's in R)

In this exercise, you will construct in R many distinct realizations of confidence intervals for an unknown mean, and return the fraction of intervals containing the true mean.

Let $X_1, ..., X_n \overset{iid}{\sim} \mathrm{N}(\mu, \sigma^2)$, where the value of μ is known, and σ^2 is assumed to be known.

(a) In R, write a piece of code that does the following:

INPUTS:

- Population mean μ (real number)
- Population variance σ^2 (positive real number)
- Sample size n (positive integer)
- Confidence level α (in $(0, 1)$)
- Number of trials M (positive integer)

PROCEDURE:

- For each trial in $1, 2, ..., M$:
  - Generate realizations $x_1, ..., x_n$ from a $\mathrm{N}(\mu, \sigma^2)$ distribution.
  - Use $x_1, ..., x_n$ to compute a realization of the two-sided $1 - \alpha$ confidence interval for the population mean, given by $\bar{x}_n \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$.
    Note: for the critical values of a standard normal distribution, use the `qnorm()` function.
  - Check whether this interval contains μ.
- Return the fraction of the M trials in which the realized $1 - \alpha$ confidence interval contained the true population mean.

Note: For this part of the exercise, you need only submit your code – a screenshot will suffice.

(b) Let $\mu = \frac{1}{2}$, $\sigma^2 = \frac{1}{4}$, $n = 10$, $\alpha = 5\%$, and $M = 1000$, and run your procedure. Report the fraction of confidence intervals that contained the true mean. What do you see? Why?

(c) Repeat the same exercise, but with $n = 100$. What, if anything, changes? Why?

Exercise: 2. (CI's for Uniform Distribution Parameter)
Let $X_1, X_2, \ldots, X_n \overset{iid}{\sim} \text{Unif}[0, \theta]$, each of whose density is given by:

$$f(x;\theta) = \begin{cases} \frac{1}{\theta} & 0 \le x \le \theta \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y := \frac{\max_{i=1\ldots n} X_i}{\theta}$, whose density is given by:

$$f_Y(y) = \begin{cases} ny^{n-1} & 0 \le y \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Use $f_Y(y)$ to verify that

$$\mathbb{P}\left((\alpha/2)^{1/n} \le Y \le (1-\alpha/2)^{1/n}\right) = 1 - \alpha$$

and use this statement to build a $1-\alpha$ confidence interval for θ.

Hint: verify the statement by computing the CDF of Y.

(b) Verify that

$$\mathbb{P}\left(\alpha^{1/n} \le Y \le 1\right) = 1 - \alpha,$$

and build a $1-\alpha$ confidence interval for θ based on this statement.

Exercise: 3. (CI's for Exponential Distribution Parameter)
Let $X_1, ..., X_n \overset{iid}{\sim} \text{Exp}(\lambda)$, for some unknown parameter $\lambda > 0$. Assume that $n > 40$. We will build two distinct $1 - \alpha$ confidence intervals for the parameter λ.

(a) Let $\bar{X}_n := \frac{1}{n} \sum_{i=1}^n X_i$ and $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.
Show that for any given $\alpha \in (0, 1)$:
$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\sqrt{n}\left(\bar{X}_n - \frac{1}{\lambda}\right)}{s} \leq z_{\alpha/2}\right) = 1 - \alpha,$$
where $z_{\alpha/2}$ is the $\alpha/2$ critical value for a standard normal random variable.

(b) Use the expression in part (a) to construct a two-sided $1 - \alpha$ confidence interval for λ. Be careful to notice that λ is **not** the mean of $X_1, ..., X_n$.

(c) Build a **lower** one-sided $1 - \alpha$ confidence interval for λ. Be sure to begin with a probabilistic statement (as in part (a)), justify it, and show your work in obtaining the confidence interval. Your interval for λ should be of the form $[\cdot, \infty)$.

Exercise: 4. (Setting up Hypothesis Tests)
Let $X_1, ..., X_n$ be iid random variables with mean μ and variance σ^2. Moreover, let $\bar{X}_n$ be the (random) sample mean and $s^2(X_1, ..., X_n)$ be the (random) sample standard deviation.

For each of the following assertions, state whether it is a legitimate set of hypotheses. If they are not legitimate hypotheses, justify why not.

(a) $H_0 : \mu = 1$ and $H_A : \mu > 1$.

(b) $H_0 : \frac{\mu}{\sigma} = 1$ and $H_A : \frac{\mu}{\sigma} > 1$.

(c) $H_0 : \bar{X}_n = 1$ and $H_A : \bar{X}_n > 1$.

(d) $H_0 : \frac{\bar{X}_n - \mu}{s/\sqrt{n}} = 1$ and $H_A : \frac{\bar{X}_n - \mu}{s/\sqrt{n}} > 1$.

Exercise: 5. [OPTIONAL] (Two-Sample Confidence Intervals)
Note: This is an ***optional*** exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let $X_1, ..., X_n \overset{iid}{\sim} N(\mu_X, \sigma_X^2)$ and $Y_1, ..., Y_n \overset{iid}{\sim} N(\mu_Y, \sigma_Y^2)$, such that all of the random variables $X_1, ..., X_n, Y_1, ..., Y_n$ are mutually independent.

(a) What is the distribution of the difference $X_1 - Y_1$? Be sure to state and justify: i) the family of distribution, and ii) any relevant parameters.
Hint: Recall that all of these random variables are mutually independent.

(b) What is the distribution of the *average* difference $\frac{1}{n} \sum_{i=1}^n (X_i - Y_i)$?

(c) Use the result in part (b) to build a two-sided $1 - \alpha$ confidence interval for the *difference* in means, $\mu_X - \mu_Y$. Be sure to: i) start with a justified probabilistic statement, and ii) show every step in deriving your confidence interval.